

BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

Final Internship Capstone Project Report

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Technologies Used

Python • Pandas • NumPy • Matplotlib
Power BI • Git & GitHub • Jupyter Notebook

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The objective of this project is to build a comprehensive Mutual Fund Analytics Platform using Python, Power BI, and financial analytics techniques. The project aims to analyze mutual fund performance, investor behaviour, portfolio risk, and market trends through an end-to-end data analytics pipeline.

Chapter 1

Introduction

1.1 Background

The Indian mutual fund industry has experienced significant growth over the past decade due to increasing financial awareness, digital investment platforms, and systematic investment plans (SIPs). Investors today have access to thousands of mutual fund schemes across multiple Asset Management Companies (AMCs), making fund selection and performance comparison increasingly challenging.

With the rapid expansion of investment options, investors require analytical tools that not only compare returns but also evaluate investment risk, portfolio diversification, and investor behaviour. Data Analytics and Business Intelligence techniques provide an effective solution by converting raw financial data into meaningful insights for informed investment decisions.

This project presents an end-to-end Mutual Fund Analytics Platform developed using Python and Power BI. The platform integrates data engineering, financial analytics, advanced risk metrics, investor analytics, and interactive dashboards to provide comprehensive insights into mutual fund performance.

1.2 Problem Statement

Investors often face difficulties while selecting mutual funds because information is scattered across multiple sources and traditional performance measures do not adequately represent investment risk.

The major challenges include:

- Difficulty comparing mutual fund schemes across different AMCs.
- Limited understanding of portfolio risk and downside exposure.
- Lack of integrated dashboards for performance monitoring.

- Difficulty identifying suitable funds based on investor risk appetite.
- Limited analysis of investor behaviour and SIP continuity.

Therefore, there is a need to develop a centralized analytics platform capable of processing historical mutual fund data and generating meaningful investment insights.

1.3 Project Objectives

The primary objectives of this project are:

- Develop a complete ETL pipeline for mutual fund datasets.
- Clean and preprocess raw financial datasets.
- Perform Exploratory Data Analysis (EDA).
- Evaluate mutual fund performance using financial metrics.
- Compute advanced risk measures including VaR and CVaR.
- Analyse investor behaviour using cohort analysis.
- Develop a rule-based mutual fund recommendation system.
- Evaluate portfolio diversification using the Herfindahl-Hirschman Index (HHI).
- Design an interactive Power BI dashboard.
- Generate actionable business recommendations.

1.4 Scope of the Project

The project covers the complete data analytics lifecycle for mutual fund data. It begins with data collection and preprocessing, followed by exploratory data analysis, financial performance evaluation, advanced analytics, dashboard development, and business recommendations.

The analysis includes historical NAV trends, investor transactions, scheme performance, portfolio holdings, and benchmark data. However, the project does not perform real-time market prediction or live trading recommendations.

1.5 Technology Stack

The project was developed using the following technologies:

Category	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Visualization	Matplotlib
Business Intelligence	Power BI
Development Environment	Jupyter Notebook
Version Control	Git and GitHub
IDE	VS Code

Table 1.1: Technology Stack Used in the Project

Chapter 2

Data Collection and ETL Pipeline

2.1 Dataset Overview

The project utilizes multiple datasets related to the Indian Mutual Fund industry. These datasets were obtained from the Bluestock Mutual Fund Capstone dataset and contain information regarding mutual fund schemes, daily NAV history, investor transactions, portfolio holdings, and scheme performance.

The datasets collectively provide sufficient information to perform financial analysis, investor behaviour analysis, risk assessment, and dashboard visualization.

2.2 Datasets Used

Dataset	Description
NAV History	Daily Net Asset Value (NAV) of mutual fund schemes used for return and risk calculations.
Transactions	Investor transaction records including SIP, Lumpsum investments and Redemptions.
Scheme Performance	Performance indicators such as CAGR, Sharpe Ratio, Alpha, Beta and Risk Grade.
Holdings	Portfolio allocation of each mutual fund across different sectors and companies.

Table 2.1: Datasets Used in the Project

2.3 ETL Pipeline

The project follows a structured ETL (Extract, Transform and Load) process to ensure high-quality analytical outputs.

2.3.1 Extract

The raw datasets were collected from Excel and CSV files. Multiple files containing NAV history, investor transactions, scheme details and holdings were imported into Python using the Pandas library.

2.3.2 Transform

Several preprocessing operations were performed to improve data quality.

- Missing value treatment
- Duplicate record removal
- Date format standardization
- Data type conversion
- Outlier identification
- Feature engineering
- Data validation

These transformations ensured consistency across all datasets before performing further analysis.

2.3.3 Load

The cleaned datasets were stored in the processed data folder and used throughout the project for Exploratory Data Analysis (EDA), Performance Analytics, Risk Analysis, and Power BI Dashboard development.

2.4 Data Cleaning

The following cleaning operations were performed:

- Removed duplicate records.
- Handled missing values appropriately.
- Converted date columns into datetime format.
- Standardized numerical columns.
- Corrected inconsistent categorical values.

- Validated AMFI codes and scheme identifiers.
- Created derived variables for financial analysis.

These preprocessing steps significantly improved the quality and reliability of the analytical results.

2.5 Project Workflow

The overall workflow of the project is illustrated below:

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Performance Analytics
5. Advanced Risk Analytics
6. Dashboard Development
7. Business Insights and Recommendations

The workflow ensures a systematic approach from raw data acquisition to actionable business intelligence.

Chapter 3

Exploratory Data Analysis (EDA)

3.1 Overview

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the mutual fund datasets before performing financial analysis and dashboard development. The primary objective was to identify trends, missing values, distributions, correlations, and investor behaviour patterns.

The EDA process helped in validating the data quality and discovering useful insights that were later incorporated into the analytical models and Power BI dashboard.

3.2 Data Quality Assessment

Before analysis, the datasets were examined for data quality issues.

The following preprocessing checks were performed:

- Missing value analysis
- Duplicate record detection
- Data type verification
- Date format validation
- Numerical range checking
- Categorical value consistency

The cleaning process ensured that all datasets were suitable for further financial analysis.

3.3 NAV Trend Analysis

The historical Net Asset Value (NAV) data was analysed to understand the growth pattern of mutual fund schemes over time.

Key observations include:

- Most equity-oriented schemes showed a consistent upward NAV trend over the long term.
- Short-term fluctuations reflected market volatility.
- NAV history served as the basis for daily return calculation and risk analysis.

3.4 Investor Transaction Analysis

Investor transaction records were analysed to understand investment behaviour.

The dataset contained three major transaction types:

- SIP (Systematic Investment Plan)
- Lumpsum Investment
- Redemption

The analysis indicated that SIP transactions formed the largest proportion of total transactions, highlighting investors' preference for disciplined long-term investing.

3.5 Investor Demographics

Investor demographics were analysed using age, gender, and city information.

Important observations include:

- Male investors represented the majority of investors.
- The 26–35 years age group contributed the highest number of investors.
- Tier-1 (T30) cities accounted for a larger share of investments than B30 cities.

These findings indicate that mutual fund participation is highest among young working professionals living in urban regions.

3.6 Fund Performance Distribution

The scheme performance dataset was analysed using important financial indicators.

The following metrics were examined:

- CAGR
- Sharpe Ratio
- Alpha
- Beta
- Risk Grade

The analysis revealed variation in fund performance across different schemes, providing the basis for later recommendation and ranking models.

3.7 Category-wise Analysis

Mutual fund schemes were further analysed based on their investment category.

The dashboard indicated that:

- Liquid Funds received the highest net inflows.
- Sectoral/Thematic Funds showed strong investor interest.
- Large and Mid Cap Funds maintained consistent investment growth.

This demonstrates a balanced investor preference between low-risk liquidity-focused funds and high-growth equity funds.

3.8 Key EDA Findings

The exploratory analysis produced several important findings:

- Mutual fund industry assets continue to grow steadily.
- SIP remains the dominant investment mode.
- Young investors (26–35 years) contribute the highest participation.
- Large AMCs dominate industry assets under management.
- Equity funds generally exhibit higher long-term returns along with higher volatility.
- Data quality after preprocessing was suitable for advanced financial analytics.

Chapter 4

Performance Analytics

4.1 Overview

Performance Analytics was conducted to evaluate the historical performance of mutual fund schemes using standard financial indicators. The analysis focused on measuring returns, risk-adjusted performance, and fund consistency across different market conditions.

The primary objective was to identify high-performing mutual funds and compare them using quantitative financial metrics.

4.2 Performance Metrics Used

The following performance indicators were used throughout the project:

Metric	Purpose
CAGR	Measures long-term annualized return of a mutual fund.
Sharpe Ratio	Measures risk-adjusted return by comparing excess return with volatility.
Alpha	Indicates the fund's excess return over its benchmark.
Beta	Measures sensitivity of the fund relative to market movements.
Rolling Sharpe Ratio	Evaluates consistency of risk-adjusted performance over time.

Table 4.1: Financial Performance Metrics

4.3 Compound Annual Growth Rate (CAGR)

CAGR represents the average annual growth rate of a mutual fund over a specified period.

It provides a standardized measure for comparing funds irrespective of investment duration.

Higher CAGR values generally indicate stronger long-term performance.

4.4 Sharpe Ratio Analysis

The Sharpe Ratio evaluates the return generated per unit of risk undertaken.

Funds with higher Sharpe Ratios provide better risk-adjusted performance and are generally preferred by investors seeking efficient returns.

The dashboard indicates an average Sharpe Ratio of approximately 0.54 across the analysed schemes.

4.5 Alpha and Beta Analysis

Alpha measures whether a mutual fund outperformed its benchmark, while Beta measures the volatility of the fund relative to the overall market.

The analysis revealed that several equity-oriented schemes generated positive Alpha while maintaining acceptable Beta values, indicating efficient portfolio management.

4.6 Rolling Sharpe Ratio

To evaluate consistency over time, a 90-day Rolling Sharpe Ratio was computed for selected mutual fund schemes.

The rolling analysis helped identify periods of stable performance as well as intervals of increased market volatility.

Funds maintaining consistently positive Rolling Sharpe Ratios demonstrated superior long-term stability.

4.7 Fund Ranking

Mutual funds were ranked based on multiple financial indicators including CAGR, Sharpe Ratio, Alpha, Beta, and composite fund score.

The ranking process enabled quick identification of high-performing schemes suitable for different investor profiles.

Top-performing funds generally exhibited:

- Higher CAGR
- Higher Sharpe Ratio

- Positive Alpha
- Moderate Beta
- Lower downside risk

4.8 Performance Insights

Based on the analysis, the following observations were made:

- Equity mutual funds generated higher long-term returns than debt-oriented schemes.
- Funds with higher Sharpe Ratios consistently delivered superior risk-adjusted performance.
- Rolling Sharpe analysis highlighted performance stability across different market periods.
- Large-cap funds demonstrated relatively lower volatility compared to small-cap funds.
- The ranking model effectively differentiated strong-performing and weak-performing schemes.

4.9 Business Interpretation

Performance Analytics provides valuable information for both investors and fund managers.

Investors can use these metrics to select schemes matching their investment objectives and risk tolerance, while fund managers can benchmark portfolio performance against industry peers and identify opportunities for portfolio optimization.

Chapter 5

Advanced Analytics and Risk Metrics

5.1 Overview

Advanced Analytics was performed to evaluate portfolio risk, investor behaviour, and investment suitability beyond traditional performance metrics. The analysis incorporated statistical risk measures, investor segmentation techniques, portfolio concentration analysis, and a rule-based recommendation system.

The objective was to provide actionable insights that assist investors in making informed investment decisions while understanding the associated risks.

5.2 Value at Risk (VaR)

Value at Risk (VaR) estimates the maximum expected loss in a portfolio over a specified confidence level.

In this project, Historical VaR was calculated using the 5th percentile of the daily return distribution for each mutual fund.

Mathematically,

$$VaR_{95} = 5^{th} \text{ percentile of daily returns}$$

A lower (more negative) VaR indicates greater downside risk.

The analysis showed that equity-oriented funds generally exhibited higher VaR due to greater market volatility, while comparatively stable funds demonstrated lower downside exposure.

5.3 Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR), also known as Expected Shortfall, measures the average loss beyond the VaR threshold.

It is calculated as:

$$CVaR = Mean(Returns < VaR)$$

CVaR provides a better understanding of extreme downside risk by considering the severity of losses occurring beyond the Value at Risk limit.

Funds with highly negative CVaR values indicate greater vulnerability during adverse market conditions.

5.4 Rolling Sharpe Ratio Analysis

A rolling 90-day Sharpe Ratio was calculated for selected mutual fund schemes to analyse performance consistency.

The rolling analysis captured changes in risk-adjusted returns over time and highlighted periods of stable performance as well as increased market volatility.

Consistently positive rolling Sharpe Ratios indicate efficient portfolio management and stable long-term performance.

5.5 Investor Cohort Analysis

Investor Cohort Analysis was performed by grouping investors according to the year of their first investment transaction.

The analysis measured:

- Average SIP Amount
- Total Investment
- Preferred Mutual Fund Schemes

The results indicated that investors entering during 2024 contributed the highest overall investment volume, while the 2025 cohort represented newer investors with comparatively lower accumulated investments.

Cohort analysis enables fund houses to understand investor acquisition trends and evaluate long-term customer engagement.

5.6 SIP Continuity Analysis

Systematic Investment Plan (SIP) continuity was analysed to identify investors at risk of discontinuing their investments.

For investors with at least six SIP transactions, the average interval between successive SIPs was calculated.

Investors having an average gap greater than 35 days were classified as **At-Risk**.

This analysis can help Asset Management Companies proactively engage investors who are likely to discontinue their SIPs.

5.7 Fund Recommendation System

A simple rule-based recommendation model was developed to recommend mutual fund schemes according to investor risk appetite.

Investors were classified into three categories:

- Low Risk
- Moderate Risk
- High Risk

The recommendation engine filtered schemes based on the Risk Grade and ranked them using the Sharpe Ratio.

For Moderate Risk investors, the recommendation model identified high-performing large-cap funds offering attractive risk-adjusted returns.

This model demonstrates how financial analytics can support personalized investment decision-making.

5.8 Sector Concentration Analysis

Portfolio diversification was evaluated using the Herfindahl-Hirschman Index (HHI).

The HHI was computed as:

$$HHI = \sum_{i=1}^n w_i^2$$

where w_i represents the proportion of investment allocated to each sector.

Interpretation:

- Lower HHI indicates better diversification.
- Higher HHI indicates greater portfolio concentration.

The analysis revealed that a few equity funds maintained relatively high HHI values, indicating concentrated exposure to a limited number of sectors.

Diversified portfolios generally reduce unsystematic investment risk.

5.9 Key Findings

The Advanced Analytics module generated several important insights:

- Historical VaR and CVaR successfully quantified downside investment risk.
- Rolling Sharpe Ratio highlighted performance consistency over time.
- Investor Cohort Analysis identified stronger investment activity among earlier cohorts.
- SIP Continuity Analysis helped identify investors at risk of discontinuing investments.
- The recommendation system successfully matched investors with suitable mutual fund schemes based on their risk profile.
- HHI analysis identified funds with high sector concentration that may require greater diversification.

5.10 Business Significance

The Advanced Analytics framework extends traditional mutual fund analysis by combining financial risk assessment, behavioural analytics, and investment recommendations.

These analytical techniques can support investors, financial advisors, and Asset Management Companies in making data-driven investment decisions, improving customer retention, and managing portfolio risk more effectively.

Chapter 6

Power BI Dashboard and Business Insights

6.1 Dashboard Overview

An interactive Power BI dashboard was developed to transform the analytical outputs into meaningful business insights. The dashboard enables users to explore mutual fund performance, investor behaviour, industry trends, and portfolio characteristics through interactive visualizations.

The dashboard consists of four pages covering industry overview, fund performance, investor analytics, and market trends.

6.2 Dashboard Features

The dashboard provides the following functionalities:

- Industry-level Mutual Fund Overview
- Asset Management Company (AMC) Comparison
- Fund Performance Analysis
- Risk vs Return Visualization
- Investor Demographic Analysis
- SIP Trend Analysis
- Category-wise Fund Flow Analysis
- Interactive Filtering and Drill-down

These visualizations allow users to analyse mutual fund data from multiple perspectives and support informed investment decisions.

6.3 Industry Overview

The dashboard indicates that the Indian Mutual Fund industry has experienced strong growth over recent years.

Major observations include:

- Assets Under Management (AUM) of the top AMCs increased significantly, reflecting sustained industry expansion.
- Monthly SIP inflows crossed Rs.31,000 crore, highlighting increasing retail participation.
- The industry consists of more than 26 crore investor folios and multiple mutual fund schemes across different categories.
- Rising SIP contributions indicate increasing investor confidence in long-term wealth creation.

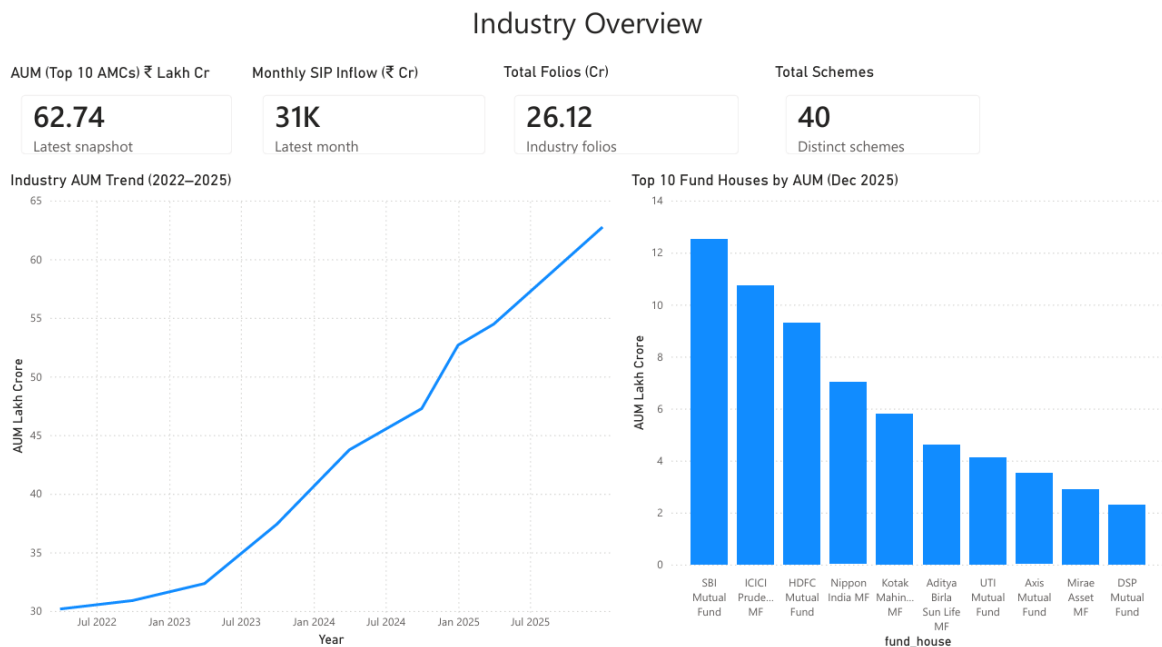


Figure 6.1: Dashboard Page 1: Industry Overview

6.4 Asset Management Company (AMC) Analysis

The dashboard compares the Assets Under Management of leading Asset Management Companies.

Key findings include:

- SBI Mutual Fund maintains the highest Assets Under Management.
- ICICI Prudential Mutual Fund and HDFC Mutual Fund rank among the leading fund houses.
- A significant proportion of industry assets is concentrated among a few large AMCs.
- Smaller AMCs continue to expand but contribute comparatively lower market share.

This demonstrates a highly competitive market with strong dominance by established fund houses.

6.5 Mutual Fund Performance Analysis

The dashboard evaluates scheme performance using financial metrics such as CAGR, Sharpe Ratio, and Fund Score.

The analysis indicates:

- Average Sharpe Ratio is approximately 0.54.
- Average CAGR is approximately 16.41%.
- Several large-cap funds consistently achieved higher fund scores.
- Better-performing funds generally combined strong returns with moderate risk.

The Risk-Return scatter plot illustrates a positive relationship between CAGR and Sharpe Ratio, indicating that funds with superior risk-adjusted performance also tend to generate stronger long-term returns.

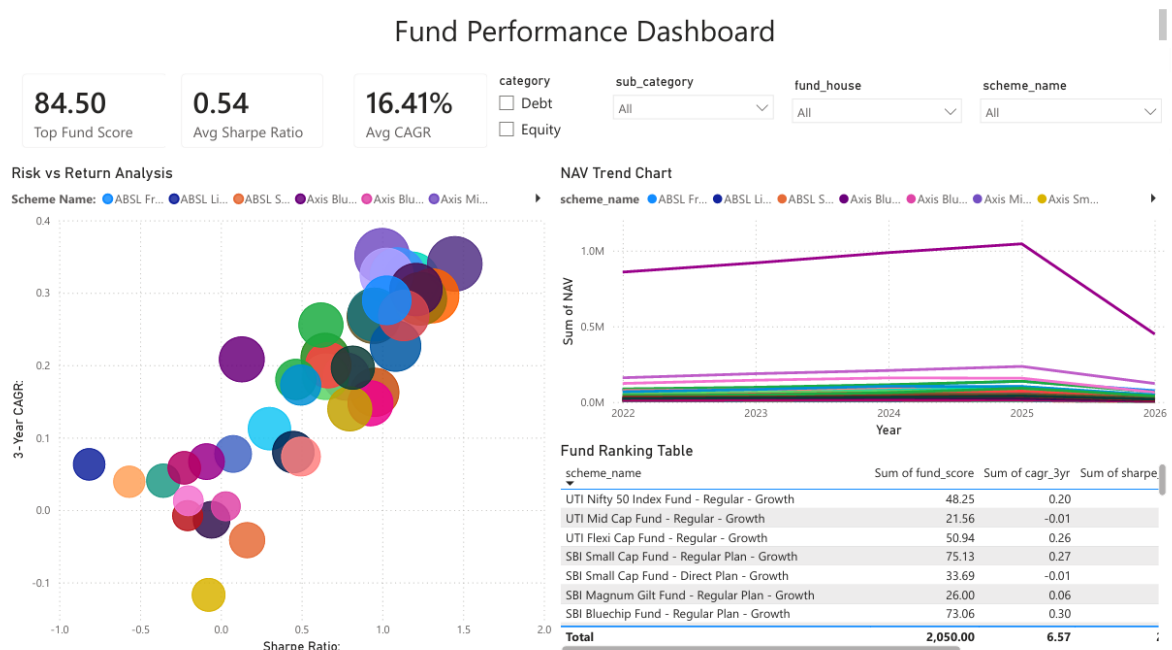


Figure 6.2: Dashboard Page 2: Investor Analytics

6.6 Investor Demographics

Investor demographic analysis highlights participation across different age groups, gender categories, and city classifications.

The dashboard reveals:

- Male investors account for approximately 66.5% of the investor base.
- Female investors represent approximately 33.5%.
- Investors aged between 26 and 35 years constitute the largest investor segment.
- Tier-1 (T30) cities contribute a significantly larger share of investments than B30 cities.

These findings suggest that mutual fund investments are particularly popular among young urban professionals.

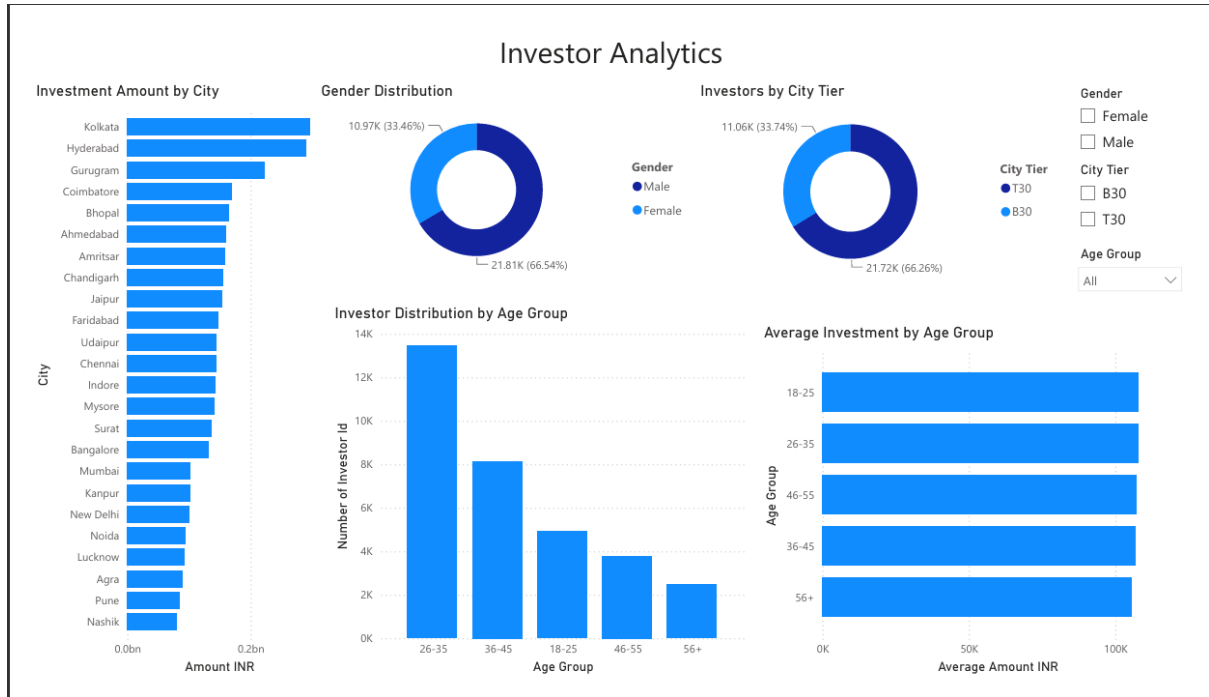


Figure 6.3: Dashboard Page 3: Investor Analytics

6.7 Category-wise Analysis

Category-wise fund flow analysis provides insights into investor preferences across different mutual fund categories.

Important observations include:

- Liquid Funds recorded the highest net inflows.
- Sectoral and Thematic Funds attracted substantial investor interest.
- Large and Mid Cap Funds continued to receive consistent investments.
- Investors demonstrated a balanced preference between liquidity-oriented and growth-oriented investment strategies.

6.8 SIP Trend Analysis

The dashboard illustrates a steady increase in Systematic Investment Plan (SIP) participation.

Key observations include:

- SIP Accounts increased substantially between 2022 and 2025.
- SIP Assets Under Management reached approximately 15.9 lakh crore.

- Monthly SIP contributions continued to exhibit positive growth.
- The highest year-on-year growth exceeded 50%, indicating strong retail investor confidence.

The increasing adoption of SIPs highlights the growing awareness of disciplined long-term investing among retail investors.

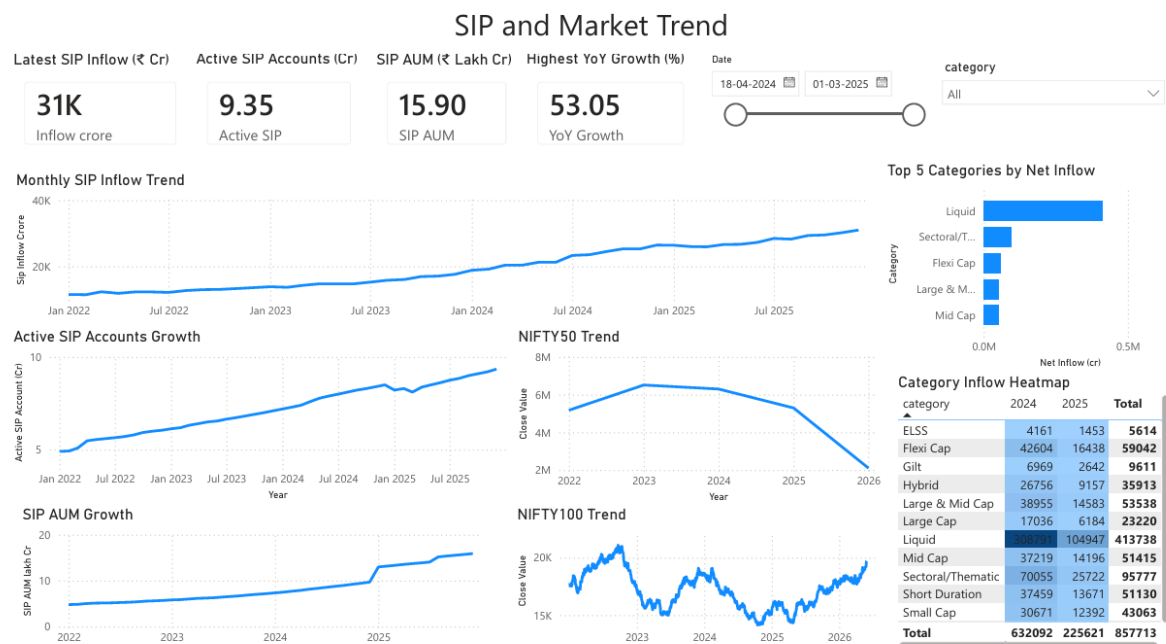


Figure 6.4: Dashboard Page 4: Market Trends and SIP Analysis

6.9 Business Insights

The dashboard generated several valuable business insights:

- India's mutual fund industry continues to expand rapidly due to increasing retail participation.
- Large Asset Management Companies dominate the market in terms of Assets Under Management.
- Young investors represent the fastest-growing investor segment.
- Equity-oriented funds generally deliver higher long-term returns while carrying relatively higher investment risk.
- SIP remains the preferred investment mode for long-term wealth creation.
- Interactive dashboards significantly improve investment monitoring and decision-making.

6.10 Detailed Dashboard Insights

Based on the Power BI dashboard, the following significant business insights were obtained:

- The Indian Mutual Fund industry is experiencing rapid growth, with Assets Under Management (AUM) of the top Asset Management Companies increasing from approximately 30 lakh crore in 2022 to over 62 lakh crore in the latest period.
- Monthly SIP inflows have crossed Rs. 31,000 crore, indicating increasing retail investor participation and growing confidence in long-term wealth creation.
- The industry currently serves more than 26 crore investor folios across multiple mutual fund schemes, demonstrating widespread adoption of mutual fund investments.
- SBI Mutual Fund holds the highest Assets Under Management, followed by ICICI Prudential Mutual Fund and HDFC Mutual Fund, indicating strong market concentration among leading fund houses.
- The average Sharpe Ratio across analysed schemes is approximately 0.54, while the average CAGR is around 16.41%, suggesting satisfactory long-term risk-adjusted performance.
- The Risk-Return analysis demonstrates a positive relationship between Sharpe Ratio and CAGR, indicating that better risk-adjusted funds generally generate superior long-term returns.
- Male investors constitute approximately 66.5% of the investor base, whereas female investors account for approximately 33.5%.
- Investors aged between 26 and 35 years represent the largest investment segment, highlighting the growing participation of young professionals in mutual fund investments.
- Tier-1 (T30) cities contribute nearly two-thirds of total investments, indicating higher financial awareness and investment activity in metropolitan regions.
- Liquid Funds recorded the highest net inflows, followed by Sectoral/Thematic Funds and Large & Mid Cap Funds, reflecting a balanced investor preference towards both liquidity and long-term capital appreciation.
- SIP Accounts have increased substantially between 2022 and 2025, while SIP Assets Under Management have continued to register consistent year-on-year growth.

- Interactive dashboard visualizations enable investors and fund managers to identify high-performing schemes, monitor market trends, and make data-driven investment decisions efficiently.

6.11 Dashboard Benefits

The Power BI dashboard provides several practical advantages:

- Simplifies complex financial information.
- Supports data-driven investment decisions.
- Enables quick comparison of mutual fund schemes.
- Helps identify high-performing and high-risk funds.
- Improves investor awareness through interactive visualizations.
- Assists Asset Management Companies in monitoring customer behaviour and market trends.

Chapter 7

Recommendations, Limitations and Conclusion

7.1 Recommendations

Based on the analytical findings and dashboard insights, the following recommendations are proposed for investors, Asset Management Companies (AMCs), and financial advisors.

7.1.1 Recommendations for Investors

- Select mutual funds based on both returns and risk-adjusted performance rather than returns alone.
- Consider investing through Systematic Investment Plans (SIPs) to benefit from disciplined investing and rupee cost averaging.
- Diversify investments across multiple sectors and fund categories to reduce concentration risk.
- Monitor portfolio performance periodically using metrics such as Sharpe Ratio, VaR, and CAGR.
- Choose mutual fund schemes according to individual risk appetite instead of following market trends.

7.1.2 Recommendations for Asset Management Companies

- Use investor cohort analysis to improve customer acquisition and retention strategies.
- Identify investors with irregular SIP contributions and initiate engagement campaigns before SIP discontinuation.

- Improve portfolio diversification in highly concentrated funds to reduce investment risk.
- Develop personalized fund recommendation systems using investor profiles and historical investment behaviour.
- Utilize interactive dashboards for continuous monitoring of fund performance and investor trends.

7.2 Limitations of the Study

Although the project provides meaningful insights, certain limitations exist.

- The analysis is based on historical mutual fund data and may not accurately predict future market performance.
- Live market prices and real-time NAV updates were not integrated into the analytical models.
- The recommendation system follows a rule-based approach and does not utilize Machine Learning algorithms.
- Portfolio analysis is limited to the available holdings dataset.
- External economic indicators such as inflation, interest rates, and geopolitical events were not incorporated into the analysis.

7.3 Future Scope

Several enhancements can further improve the capabilities of this Mutual Fund Analytics Platform.

- Integration of live mutual fund NAV data using financial APIs.
- Development of Machine Learning models for mutual fund return prediction.
- AI-powered personalized investment recommendation engine.
- Portfolio optimization using Modern Portfolio Theory (MPT).
- Real-time risk monitoring dashboards.
- Mobile application for investor portfolio tracking.
- Automated report generation and alert notifications.

7.4 Conclusion

The Bluestock Mutual Fund Analytics Platform successfully demonstrates the application of Data Analytics and Business Intelligence techniques within the mutual fund industry.

The project integrates the complete analytics lifecycle, beginning with data collection and preprocessing, followed by Exploratory Data Analysis, financial performance evaluation, advanced risk analytics, investor behaviour analysis, and interactive dashboard development.

Advanced financial metrics such as Sharpe Ratio, Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Fund Recommendation, and Sector Concentration Analysis provide deeper insights into investment performance and portfolio risk.

The Power BI dashboard transforms complex financial information into interactive visualizations that enable investors and financial professionals to monitor fund performance, understand investor behaviour, and make informed investment decisions.

Overall, the project demonstrates how Data Analytics, Python, and Business Intelligence tools can support data-driven decision making in the financial services industry and create value for investors as well as Asset Management Companies.

7.5 Learning Outcomes

During the completion of this project, the following technical and analytical skills were strengthened:

- Python Programming
- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Financial Performance Analysis
- Advanced Risk Analytics
- Power BI Dashboard Development
- Business Intelligence and Data Visualization
- Git and GitHub Version Control
- Report Writing and Technical Documentation

7.6 References

The following resources were referred to during the development of this project.

1. Association of Mutual Funds in India (AMFI).
2. Securities and Exchange Board of India (SEBI).
3. Pandas Documentation.
4. NumPy Documentation.
5. Matplotlib Documentation.
6. Microsoft Power BI Documentation.
7. Python Official Documentation.
8. Bluestock Mutual Fund Capstone Dataset.